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## *Instruction*

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## Abstract

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# Chapter 1

# Introduction

## 1.1 Phishing as an Escalating Cybersecurity Threat[prp]

This paper will present a comprehensive survey of the phishing email detection systems landscape, with a specific emphasis on Multi-Layered Defensive Architectures with Quantum-Enhancement in the Era of AI-Boosted Phishing techniques. While phishing detection has been widely studied, NLP-centric approaches – especially semantic and deep learning-based models – remain underexplored, fragmented, and insufficiently benchmarked. The paper positions phishing emails as a persistent, escalating cybersecurity threat that traditional defensive countermeasures are actively exploited and bypassed. It remains as the most economically expensive forms of cybercrime. It will also address attacker's techniques used in bypassing these measures. Despite decades of defensive academic research, it remains as the most successful attack vector for attackers [1], [2].

Phishing is defined as a social engineering-driven offensive vector that deceives computer users into revealing sensitive information (credentials, banking

data, identity details) primarily via email (or could be SMS, Voice, Video?) as the dominant attack surface/vector due to its scalability, low operational cost, and high trust inheritance within organizational workflows [2]. The email attack surface witnessed an exponential growth especially during crises (e.g., COVID-19, Financial Crashes?, Wars?, APTs, Political Events, Business Investments, Banking Strategies, State-Sponsored APTs) due to rapidly exploiting fear, uncertainty, and institutional disruption (APWG reports cited in [2]). The persistence of phishing is economically significant: handling a single phishing incident requires substantial labor, response time, and remediation costs, which scale dramatically at enterprise levels [1]. As emphasized in the literature, phishing is not a static threat but an adaptive ecosystem that continuously evolves to bypass deployed defenses [3], [2]. "On average, dealing with the threat of a single phishing email takes 27.5 minutes at a cost of $31.32 per phishing message. Some organization are taking much longer and paying more per phishing message, and no organization sees only a single phishing attack in any given time period" [translate it into MONEY, MONEY, MONEY (Attacks Costs vs Defensive Costs) Table Data]. "Phishing is expected to get more sophisticated and better able to evade detection over time". "Phishing is spreading beyond email".

The dual nature of phishing:

- Technical deception (malicious links, malware, spoofed URLS, attachments, Embedded HTML)
- Psychological manipulation (urgency, fear, authority, trust, spear phishing, whaling)

# Chapter 2

# Literature Review

[paraphrase ]A major challenge identified is the lack of a universally accepted definition of phishing [2], with existing definitions varying in:

1. Target (credentials vs. broader attacker benefit)
2. Attack medium (email, websites, malware)
3. Attacker impersonation scope (third parties vs. direct deception)

This definitional diversity reflects phishing's dual nature. First, it incorporates technical deception, including malicious URLs, spoofed domains, malware attachments, embedded scripts, and HTML obfuscation. Second, it relies heavily on psychological manipulation, exploiting urgency, fear, authority, trust, and personalization through spear-phishing and whaling campaigns [2], [4]. Effective detection therefore requires defenses capable of addressing both technical artifacts and linguistic intent—an aspect inadequately handled by many existing systems.

### 2.0.1 History of Phishing

The history and evolution of phishing predates to 1995, where first recorded use of social engineering was used target AOL accounts [1], [4]. It got evlolved from basic credential theft to sophisticated attacks such as: online fraud, e-commerce impersonation, Man-in-the-Browser (MitB) attacks. Modern phishing now targets individual users, employees, organizations, vendors, supply chain, and service providers, often using personalized data within the enterprise domain (spear-phishing, whaling).

Existing phishing detection approaches

The literature review categorizes phishing detection systems into three broad families:

1. Blacklist-Based Detection
    - Relies on known malicious senders, URLs, or domains.
    - Strengths: simple, fast
    - Reactive rather than proactive
    - Requires manual reporting
    - Ineffective against zero-day phishing attacks
2. Traditional Machine Learning (ML)
    - Uses classifiers such as: SVM, Random Forest, Decision Trees, Logistic Regression, Naïve Bayes
    - Depends on manual feature engineering:
        - Email body features

  - Subject line patterns
  - URLs and scripts
  - Sender metadata
- Key limitations:
  - High dependance on surface-level text
  - Poor generalization to paraphrased/semantically altered emails
  - Labor-intensive (dataset, LLM fine-tuning) and brittle feature design

3. Deep Learning (DL)
   - Introduced to override feature engineering limitations
   - Uses CNNs, RNNs, RCNNs, word embeddings, attention mechanisms.
   - Advantages:
     - Automatic feature extraction
     - Better modeling of context and structure
   - Limitations:
     - Large data requirements
     - Training & fine-tuning complexity and costs
     - Sensitivity to missing email components (e.g., headers) [Investigate]
     - Architecture complexity

### 2.0.2 NLP-Specific Features for Phishing Detection

TABLE 2.1
Major Feature Categories [2]

| **Parameter** | **Value** |
|---|---|
| Email body features | HTML structure, suspicious phrases, embedded links. |
| Subject line features | Urgency cues (“verify”, “account”, “debit”), reply indicators |
| URL-based features | IP-based URLs, “@” symbols, redirections, external/internal link ratios. |
| Script-based features | JavaScript, pop-ups, on-click actions. |
| Sender-based features | Mismatch between sender address and reply-to address. |

### 2.0.3 NLP and ML for Phishing Email Detection

NLP and ML for Phishing Email Detection [paraphrase]

Early NLP-based phishing detection relied on [5]:

- Bag-of-Words
- N-grams
- Markov chains
- Topic models

Key problems identified:

- Over-reliance on surface text
- Inability to detect semantic rewording or intent manipulation
- Vulnerability to minor linguistic changes

Recent advances include [6], [7]:

- Word embeddings (e.g., Doc2Vec)
- Hybrid feature selection methods
- Multi-stage ML pipelines
- Deep learning architectures with attention

Some DL-based models report very high accuracy (>99%) but [2]:

- Often evaluated on limited or non-standard datasets
- Lack generalization to real-world, evolving phishing tactics
- Fail in specific edge cases (e.g., headerless emails, short URLs)

### 2.0.4 Key Gaps and Limitations Identified

The literature review consistently highlights the following unresolved issues:

- Lack of semantic understanding of email intent
- Overemphasis on links attachments, ignoring content-only phishing
- Poor handling of spear-phishing and personalized attacks [1], [2]
- Absence of standard benchmark datasets

- Limited evaluation on large, diverse, real-world corpora
- Minimal consideration of human factors (education, psychology, attacker motivation)

### 2.0.5 Discussion and Research Trends

The survey emphasizes that phishing detection must evolve from: "What features does the email contain?" to "What is the sender trying to do linguistically and psychologically?" Key future directions identified:

- Semantic NLP and intent analysis [Qwen3-Embedding-8B]
- Hybrid models combining syntax, semantics, and context
- Adaptive evaluation strategies that evolve with attacker behavior
- Bypass Techniques for the Email Protection System [Spear Phishing, Whaling Phishing, GenAI Phishing, LLM Layer Prompt Injection, LLM Denial of Service, LLM Supply Chain Vulnerabilities, LLM Manipulation, LLM Data Leakage]
- Greater focus on phisher motivation and psychology [APTs history, background, specialty, business impact & attack surface]
- Fine-tuning of LLMs per dataset quality availability
- Integration of NLP with user-awareness and behavioral defenses
- Quantum-Enhanced AI

### 2.0.6 Quantum Paradigms

Objective 4: Exploring Quantum-Enhanced Pattern Recognition in High-Dimensional Spaces Task: Evaluate the potential of Variational Quantum Classifiers (VQC) and Quantum Neural Networks (QNN) to process complex feature spaces beyond classical capabilities.

- New AI paradigms: Quantum-native algorithms may enable learning and reasoning approaches impossible on classical hardware.
- Evaluate the potential of the Enhanced Hybrid Quantum-Inspired Firefly and Artificial Bee Colony (EHQ-FABC) algorithm to achieve optimal feature selection

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